

Machine learning (ML) is a subset of artificial intelligence (AI) that develops algorithms enabling computers to learn patterns from data and make decisions or predictions without being explicitly programmed for every task. It focuses on building systems that improve their performance over time as they are exposed to more data. [[1](#), [2](#), [3](#)]

Core Types of Machine Learning

- **Supervised Learning:** Algorithms learn from labeled training data to predict outcomes (e.g., classification, regression).
- **Unsupervised Learning:** Systems find hidden patterns or structures in unlabeled data, such as clustering.
- **Reinforcement Learning:** Agents learn to make decisions by performing actions and receiving rewards or penalties.
- **Generative AI:** A branch focused on creating new content—such as text, images, or music—by learning patterns from existing data. [[1](#), [2](#)]